Network design practice 3-2

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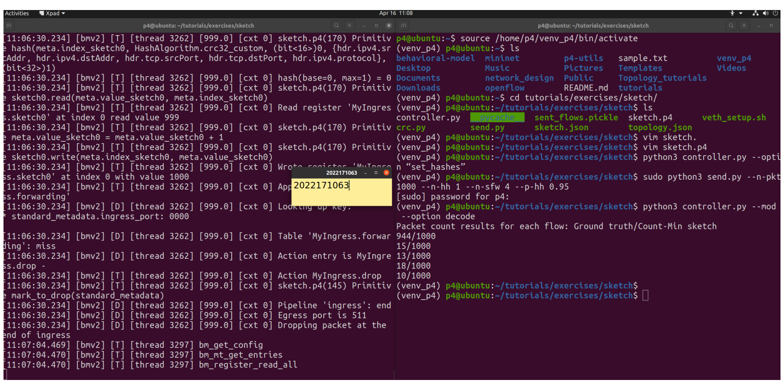
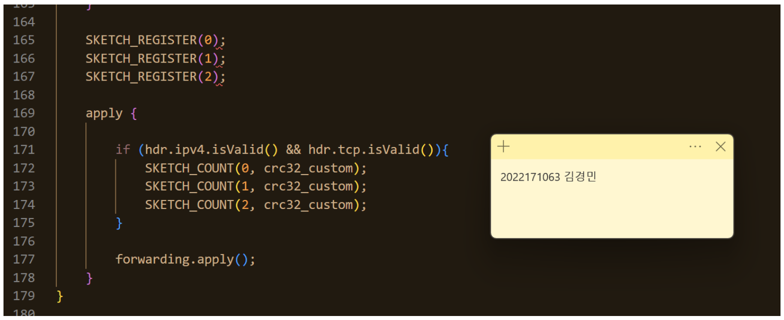
1 Assignment 1

Incrementing sketch value by 1.



After execution screen shows true count(left) and estimated count(right). Count-Min sketch never underestimates, so it estimated 1000 packets would arrive when there was 935.

2 Assignment 2

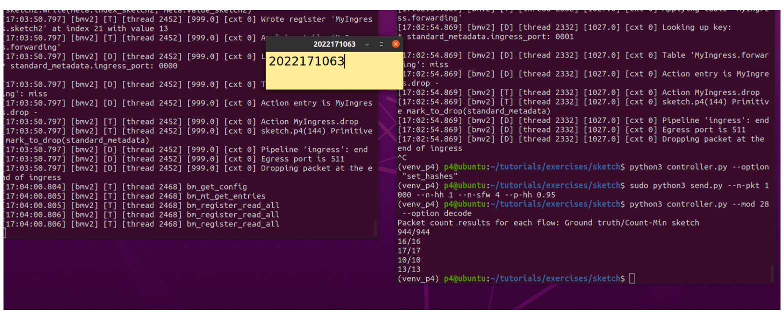


Calculated depth and width is

$$ln(1/0.05) = 3 \tag{1}$$

$$e/0.1 = 28 \tag{2}$$

I fixed the value of Sketch bucket length to 28, and made 3 sketch registers at the bottom.



Now the estimation is estimating the flow with 100% accuracy. 944 - 944

3 Assignment 3

From assignment 1, the width is 1(sketch cell bit length 1 x 8 bytes), and the depth is 1. Calculated sketch size is 1 x 1 x 8 = 8 bytes.

From assignment 2, the width is 28 and the depth is 3. Calculated sketch size is 28 x 3 x 8 = 672 bytes.

Assignment 2 required to allocate 672 bytes, which is 84 times larger compare to assignment 1 to handle heavy traffic without hash collision.

However, since it has a bigger table, the switch successfully improved accuracy by spreading the traffic accross 28 buckets and using 3 different hash functions to filter out collisions. In assignment 1, it traded off accuracy to resource usage, by cramming all traffic in 1x1 8byte box, resulting unreliable outcomes.